

Sensitivity of Estimated 10m Wind Speed to Surface Roughness Length

Roughness-length	Mean U_{10} (m s ⁻¹)	SD (m s ⁻¹)	Mean difference (m s ⁻¹)	Mean change (%)	Mean absolute difference (m s ⁻¹)	Maximum absolute difference (m s ⁻¹)	Correlation with original
Original: $z_0 = 0.02$ m	5.31	2.65	+0.00	+0.0%	0.00	0.00	1.0000
Sensitivity: $z_0 = 0.001$ m	6.10	3.04	+0.79	+14.8%	0.79	2.16	0.9995
Sensitivity: $z_0 = 0.0001$ m	6.50	3.23	+1.19	+22.3%	1.19	3.29	0.9988